

CHAPTER 7

MEIOFAUNA

CHAPTER OUTLINE AND SIGNIFICANT ELEMENTS

A Lilliputian world coexists with the macrofauna of the sedimentary habitats examined in previous chapters. This world is the realm between the sand grains: the world of meiofauna, also referred to as psammon and interstitial organisms. It was first discovered in the 20th century, and most of the early work on it was strictly taxonomic. However, recent studies have been ecological in nature. Investigations of interstitial life have been spotty with some regions remaining unexplored, whereas others have received much attention. This fact, no doubt, has more to do with the geographic distribution of researchers interested in this habitat than anything else.

This brief chapter is divided into six sections. The first section, DEFINITIONS, starts the chapter off by providing terminology. The second section, ENVIRONMENTAL CHARACTERISTICS, describes the interstitial environment. The third section, COMPOSITION OF THE INTERSTITIAL ASSEMBLAGES, presents a précis of the organisms that compose the meiobenthos. The fourth section, SAMPLING AND EXTRACTING MEIOFAUNA, discusses sampling techniques. The fifth section, ADAPTATIONS, examines life-history strategies and morphological adaptations to life between sand grains. Finally, the sixth section, ECOLOGY, presents an ecological perspective of this unique environment.

I. Definitions

Meiofaunal organisms pass through a screen of 500 μm , but are held by netting of 62 μm . Another term sometimes found in the literature that refers to the meiofauna is the psammon (G., *psammos*, sand).

endobenthos	interstitial	meiofauna
epibenthos	meiobenthos	mesobenthos

II. Environmental Characteristics

Meiofaunal organisms have been found in wet sediments from above the high-tide line to a depth of 5,000 m. The term interstitial denotes the spaces between sand grains, while meiofauna refers to the small animals that live there (G., *meio*, less; mythology, *Faunus*, god of shepherds—that is, animals).

capillary action	grain size	salinity
coarse grained	intertidal	taxonomy
eurythermal	psu	temperature
fine grained	RPD	wave action

1. Next to oxygen tension, grain size is the most important feature controlling density and diversity of interstitial organisms. Coarse-grained sediments have larger interstitial spaces, and consequently, larger organisms may colonize the habitat; with smaller grain size, just the opposite occurs. The interstitial fauna disappears when mean grain size $\leq 100 \mu\text{m}$ (0.1 mm).
2. Oxygen content is important to this habitat, and when the sediments are anoxic, the redox potential (Eh) becomes a good indicator of the species that may be present. See Chapter 6 for a discussion of the redox-potential discontinuity (RPD) layer.
3. Wave action also is an important factor influencing distribution of the psammon. If wave action is strong enough, the animals may be suspended in the plankton, where predation pressure is more severe. Besides the aforementioned elements, other factors such as temperature, salinity, and light are important to the psammon.

III. Composition of the Interstitial Assemblages

It is important that students receive a general overview of the taxonomic units composing the meiofauna. The list given next is a summary of the major forms; any additional detail is, of course, left to the discretion of the instructor.

Annelida	Gastropoda	Oligochaeta
Brachiopoda	Gastrotricha	Ostracoda
Bryozoa	Gnathostomulida	Platyhelminthes
Ciliophora	harpacticoid copepods	Polychaeta
Cnidaria	Kinorhyncha	Priapulida
Crustacea	Loricifera	Protozoa (Protists)
Echinodermata	Mollusca	Rotifera
Entoprocta	Mystacocarida	Tardigrada
flatworms	Nematoda (Nema)	Turbellaria

IV. Sampling and Extracting Meiofauna

It is not an easy thing to study these organisms. Moreover, there is no standard technique employed by meiobenthologists to study the animals that inhabit the interstitial world.

anesthetize	extraction techniques	osmotic shock
core	isotonic solution	
elutriation	MgCl ₂	

1. Coring devices, even extremely simple ones (for example, PVC tubing), are used when a quantitative sample is required; otherwise, any scoop of sediment will do to get a picture of the meiofauna. Coring devices commonly have a diameter of 2–4 cm.
2. Even if retrieving a sample is easy, processing it is not. Meiofauna has a disagreeable tendency of clinging to sand grains. Various extraction techniques have been developed to combat this problem. Some of these techniques are simple, relying on the organisms' behavioral trait of moving away from anoxic conditions. Other procedures require a specific apparatus (for example, elutriation). The use of anesthetics is often helpful, but many anesthetics are controlled substances. For more information, consult the Enrichment Exercises.

V. Adaptations

In this unit, following an untitled subsection where some general concepts are discussed, Nybakken offers a special subsection entitled "Life History." Obviously, the meiofauna comprises numerous small vermiform organisms. Often, these animals are structurally simpler than their relatives that are not members of the meiofauna.

brood protection	larvae	temporary meiofauna
cocoons	permanent meiofauna	vermiform
direct development	spermatophore	
eggs	statocysts	

1. Skeletons, spines, and other types of body armor often adorn the meiofauna. A number of good examples are seen in phylum Loricifera (*L., lorici*, armor; *L., fera*, bearing). Many forms are capable of attaching themselves to sand grains by using adhesives. In addition to being small, most meiofaunal organisms are elongate (vermiform), very flat, or both. Their small size is at times remarkable. For example, the polychaete *Diurodrilus minimus* is only as large as some protists.
2. Temporary meiofauna consist of juvenile animals that have recently settled; eventually they will outgrow the size category of meiofauna, entering the macrofaunal size. Permanent meiofaunal organisms spend their entire life in the interstitial world.
3. Being small, the meiofauna are not capable of producing large numbers of eggs. Therefore, many interstitial animals have specific adaptations to increase the probability of successful fertilization and survival of their embryos. These adaptations include spermatophores, monoecious species (hermaphrodites), and brooding of embryos.

VI. Ecology

Following an extensive introduction to this complex topic, Nybakken discusses **meiofauna–macrofauna interactions** and then **species richness, distribution, and dispersal**. By sheer numbers, the meiofauna

are overwhelming (ca. 10⁶ individuals per m²). Both density and composition vary significantly, depending on the specific environment. While their size is small, these animals still represent a respectable biomass of about 1–2 g/m².

anoxic zone	patchy	thiobios
biogeography	reduced zone	vertical migration
distribution and dispersal	species richness	zoogeography
oxidized zone	sulfide zone	

1. Factors influencing zonation patterns of the meiofauna include depth in the substratum, grain size, moisture content, oxygen concentration, and salinity. Most of the animals are found in the upper few cm. A very special assemblage (community), the thiobios, is found in anoxic layers. Most meiofaunal organisms reside in the upper sediment layers.
2. Vertical migration, a phenomenon well documented in the pelagic habitat of both freshwater and marine systems, also is seen in the interstitial habitat. The triggers for this behavior are changes in temperature and salinity.
3. Seasonal differences in the composition, abundance, and distribution of the meiofauna are well known and usually associated with the warmer months. Long-term variability has been less well studied. Predators, herbivores, detritus feeders, and suspension feeders compose the interstitial fauna.
4. Meiofauna is sensitive to disturbances, whether caused by human or by natural activities. However, meiofaunal organisms are probably not useful as indicators of pollution, because they are so difficult to work with.
5. The meiofauna have both a horizontal and a vertical distribution. Horizontal patchiness may be due to uneven distribution of resources or simply to the inability of the meiofauna to disperse.
6. Although the meiofaunal organisms are small, they are quite numerous and possess a high metabolism. Thus, total production by the group can approach that of the local macrofauna. No doubt, they represent an important factor in nutrient recycling in the sediments, and a component of the energy locked in the biomass probably makes it into higher trophic levels.
7. The biomass ratio of meiofauna to macrofauna falls in the range of 1:5 — 1:10; yet in intertidal sands and mud flats and in the deep sea, this ratio approaches 1:1.
8. The species richness of the meiofauna is notably great, in direct contrast to that of the macrofauna.
9. Many of the common meiofaunal genera appear to have a cosmopolitan distribution, but individual species may be more restricted. Lacking a dispersal stage, the meiofauna are transported as hitchhikers.

ENRICHMENT EXERCISES

Exercises with the Meiofauna

Working in the interstitial environment is very difficult, and students should learn the problems associated with it firsthand. If a marine habitat is not available, one might try sampling a freshwater one. I have found that a suitable substitute can be developed using a large freshwater or marine aquarium. However, such habitats must be started well in advance of the time for which they are needed, as it takes months, or even years, for a suitable community to develop. Be sure to keep the overlying waters well aerated; otherwise, the sediments may develop into something resembling a septic system! A sandy bottom of a depth sufficient to permit coring with PVC pipe will probably oblige your needs. At this point, a laboratory exercise may be developed in which students do their own cores, separate out the organisms by using a narcotizing agent, and observe the specimens by using the appropriate microscopes. Should your students find predators in the samples, do not have them overlook the opportunity to discover what prey parts might be present in the guts. Not only will this exercise provide some information on the species composition, but also it will document the trophic dynamics of the system.

Core Samplers

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Artificial Substrata

- Bamforth, S.S. "The variety of artificial substrates used for microfauna." In J. Cairns, Jr. (ed.). 1982. *Artificial substrates*. Ann Arbor: Ann Arbor Science.
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Elutriation Apparatus and Flotation Techniques

- Anderson, R.O. 1959. A modified flotation technique for sorting bottom fauna samples. *Limnol. Oceanogr.* 4: 223–225.
- Whitman, R.L., et al. 1983. An inexpensive and simple elutriation device for separation of invertebrates from sand and gravel. *Freshwat. Invertebr. Biol.* 2: 159–163.

Narcotizing Agents

Several of the standard narcotizing agents used by invertebrate zoologists are currently classified as controlled substances by the U.S. Drug Enforcement Administration (DEA). Instructors should take extra precautions if they use these chemicals from ancient supplies in their labs. Nevertheless, some narcotizing agents still are available. Chloral hydrate is on the DEA's list, but it is available, for example, from Sigma Chemical® if appropriate paperwork is filled out for both the DEA and Sigma. An extra charge is made by Sigma to cover its administrative costs. Instructors who wish to avoid this extra work and cost may try using magnesium sulfate, solutions of alcohol, ice water, EDTA, or oxalic acid (Lindholm, 1982) as narcotizing agents. Methanol and neosynephrin have been used with some success in freshwater systems. Also, carbon dioxide from soda water has been successful as a narcotizing agent for freshwater zooplankton; however, to my knowledge, it has not been used as an anesthetic for freshwater meiofauna.

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- Lincoln, R.J., and J.G. Sheals. 1979. *Invertebrate animals*. Cambridge, U.K.: Cambridge University Press.
- Lindholm, T. 1982. EDTA and oxalic acid—two useful agents for the narcotizing fragile and rapid microzooplankton. *Hydrobiologia* 86: 297–298.

Video

Cryptic fauna of marine sand. American Society of Zoologists. ©1992 The National Museum of Natural History (Smithsonian Institution).

For Small-Group Discussion or Laboratory Work

Nybakken notes that several taxa of the meiofauna can survive being frozen. These taxa include members of the following phyla: Platyhelminthes (Turbellaria), Gastrotricha, Annelida (Oligochaeta), and Arthropoda (Crustacea: harpacticoid copepods).

Instructors could have students discuss the following questions: What sorts of problems would these organisms have to overcome to survive the freeze–thaw cycle? What large animals can survive freezing? Of what importance is this capability to medicine? Instructors who have access to marine sediments on a routine basis could have their students develop an experimental protocol that would explore whether meiofaunal organisms can withstand freezing.

General References

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